

AGENTIC FINANCIAL OPERATIONS ASSISTANT

Team Members: (T009)

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Abstract:

Financial institutions handle a large volume of customer support requests, payment disputes, fraud investigations, and approval workflows that often require manual intervention across multiple systems, resulting in delays and operational inefficiencies. FinPilot – An AI-Powered Agentic Financial Operations Assistant is a full-stack enterprise platform developed to automate these financial operations using a multi-agent AI architecture. The system integrates specialized AI agents for ticket analysis, payment verification, fraud detection, approval management, and audit logging, enabling intelligent decision-making while ensuring human approval for high-risk actions. Built using React, Node.js, Express, PostgreSQL, and a Large Language Model (LLM), the platform operates on realistic simulated financial datasets to demonstrate real-world enterprise workflows. Every action performed by the AI is securely recorded in the database and logged for transparency, accountability, and explainability. The platform provides an interactive dashboard with real-time analytics, transaction monitoring, AI-assisted customer support, fraud analysis, refund processing, and audit management. By combining AI-driven automation with enterprise-grade workflow orchestration, FinPilot reduces manual effort, improves operational efficiency, enhances decision accuracy, and demonstrates the practical application of agentic AI in modern financial operations while maintaining security, compliance, and human oversight.

Keywords: Agentic AI, Financial Operations, FinTech, Large Language Model, Payment Verification, Fraud Detection, Refund Management, Audit Logging, React, Node.js.

ARCHITECTURE :

